GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 22 cm, $k = 2$.

44 cm

2. Area $5\,\text{cm}^2$, $k = 4$.

80 cm²

5 × 16.

3. Area $72\,\text{cm}^2$, $k = \frac{1}{3}$.

8 cm²

72 × $\frac{1}{9}$.

4. A 3 cm by 7 cm rectangle, $k = 2$. Perimeter and area of the copy.

P = 40 cm, A = 84 cm²

20 × 2; 21 × 4.

For questions 5 through 7, find k , or the ratio asked.

5. Areas $6\,\text{cm}^2$ and $150\,\text{cm}^2$. Find k .

k = 5

150 ÷ 6 = 25 = k^2.

6. Areas $50\,\text{cm}^2$ and $8\,\text{cm}^2$, original to copy. Find k .

k = 0.4

8 ÷ 50 = 0.16 = k^2, so k = 0.4: the copy is smaller.

7. Perimeters 15 cm and 45 cm, original to copy. What is the area ratio?

9

k = 3; k^2 = 9.

Choose one.

8. A figure with area $6\,\text{cm}^2$ is scaled by $k = 5$. What is the copy's area?

- ○ A. $30\,\text{cm}^2$
- ○ B. $60\,\text{cm}^2$
- ● **C.** $150\,\text{cm}^2$
- ○ D. $11\,\text{cm}^2$

C

$6 \times 5^2 = 6 \times 25$.